

Kaarigar — Build Status vs. Architecture

Where the project actually stands, measured against *Kaarigar_Architecture_v1.pdf* v1.0 and the amendments in *MIGRATION_PLAN.md*.

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Every status below was **verified against a running server on this commit**, not inferred from the plan. The single most important distinction in this report is between **an endpoint that works** and **a feature the user can reach**. The backend is substantially built; the app is almost entirely not connected to it. A status of WORKS means the API behaves correctly under test — it does not mean the demo shows it.

Status vocabulary

WORKS	Built, and verified working on this commit.
PARTIAL	Built and working at one layer, but not usable end to end.
STUBBED	Deliberately faked, labelled as such, with real architecture behind it.
NOT BUILT	Scheduled, not started.
FIX NEEDED	Built but wrong, or a known defect awaiting a decision.
MISLEADING	Returns success while doing nothing — the most dangerous state.
CUT	Removed from scope by the migration plan. Roadmap slide only.

1. Headline

The backend is roughly two days ahead of the plan. The application is not connected to it. Eighteen API endpoints are implemented and tested. The React app makes **exactly one** API call — the voice assistant — and every screen still reads from local storage. Nothing a judge sees on screen is backed by the database yet.

Area	State	Detail
Backend / API	WORKS	18 endpoints across 7 service modules, live-tested
Database	WORKS	MongoDB Atlas, read/write verified, 5 collections in use
Authentication	PARTIAL	Full JWT spine works; OTP delivery is stubbed; no login UI
Voice / AI	WORKS	Real Gemini extraction, confirmed not the keyword fallback
App data layer	NOT BUILT	All 5 screens on localStorage; 0 of 7 endpoints consumed
QR passport	NOT BUILT	The audit's #1 gap and the stated differentiator
Deployment	NOT BUILT	Config written and build-tested; never deployed

Demo loop: 5 of 10 beats are supported by working backend. Of those, only 2 are actually demonstrable in the app today (voice job entry, voice earnings entry), because the remaining three have no UI wired to them.

2. Deliberate departures from the architecture

These are decisions, not gaps. Each is recorded in *MIGRATION_PLAN.md* and should be stated openly if asked.

Architecture says	We built	Why
FastAPI (Python 3.11) backend, \$6	Express / TypeScript	A rewrite buys zero capability and costs ~2 of 10 days. The nine \$5 service boundaries are honoured exactly as router and data-access modules.

Architecture says	We built	Why
Voice is <i>Later</i> , accessibility only, \$4H	Voice is a first-class MVP input	The build-risk argument predated the code. 908 working lines already existed; cutting it would remove function and cost days.
WatermelonDB + causal-order sync, \$9	localStorage outbox	1–2 days instead of ~5, indistinguishable on stage. The append-only ledger is retained in full, so the V1 swap is a storage adapter, not a redesign.
Six roles, \$3	Two: kaarigar, customer	Only two actors are ever authenticated in the locked scope. Naming four roles we neither build nor test puts aspirational values in a frozen contract.
Redis token bucket, \$5 Tier 2	In-process rate limiter	Redis is cut. Correct for the single container we ship; must be replaced before horizontal scaling.

3. Feature modules (§4)

Module	Status	What is true today
A — Digital Passport	PARTIAL	API works: GET/PATCH /passport/me, profile auto-creates, handle allocated, ownership enforced. Public page does not exist — /p/:handle returns the React shell. The in-app QR is a Lucide icon, and Share copies a dead kaarigar .app URL.
B — Kamai Ledger	PARTIAL	API works and is genuinely append-only: idempotent writes, reversing entries, no update or delete route exists. Day/week/month summaries correct. KamaiView still reads localStorage. PDF export is still an alert().
C — Mol-Bhav Fair Price	PARTIAL	Engine works: band lookup, honest sample size, suppression below threshold, wage floor enforced server-side. Rate figures are placeholders, not sourced — see §6. No UI.
D — Reputation & Discovery	NOT BUILT	reputation-svc is an empty mounted router. No reviews, no trust score computation. Geo discovery is CUT by the plan; customers reach the worker via QR instead.
E — Job & Work Management	PARTIAL	Full 10-state machine enforced server-side — illegal transitions rejected 409, history appended, DISPUTED unreachable by design. JobsView does not display status at all and reads localStorage.
F — Financial Inclusion	CUT	V1 in the architecture. Not in the locked scope.
G — Skilling & Growth	CUT	Later in the architecture. Roadmap slide only.
H — Access Layer	PARTIAL	Multilingual UI and icon-first design work. Voice works. Offline capability NOT BUILT — the sync endpoint exists, the client outbox does not. No WhatsApp, no SMS/IVR (both cut).

4. API surface (§8)

Endpoint	Status	Notes
POST /api/auth/otp/request	STUBBED	Real challenge, hashed code, expiry, throttling. Code goes to the server log, not SMS.
POST /api/auth/otp/verify	WORKS	Real JWT, 15-min access + rotating refresh
POST /api/auth/refresh	WORKS	Rotation with reuse detection; family revoked on replay
GET/PATCH /api/passport/me	WORKS	Owner-scoped. Trust fields not self-patchable
GET/POST /api/jobs	WORKS	Idempotent on client UUIDv7
POST /api/jobs/:id/transition	WORKS	State machine enforced, concurrency-guarded
POST /api/ledger/entries	WORKS	Idempotent, append-only
GET /api/ledger/summary	WORKS	Algebraic, so reversals cancel correctly

Endpoint	Status	Notes
POST /api/ledger/entries/:id/reverse	WORKS	Original never mutated
POST /api/sync/batch	WORKS	Per-item accept/reject, causal ordering, retry policy
GET /api/pricing/band	PARTIAL	Logic correct; underlying rate data is placeholder
POST /api/assistant/process	WORKS	Real Gemini; now authenticated and rate-limited
GET /p/:handle	MISLEADING	Returns HTTP 200 with the React shell. Looks alive, serves nothing
GET /r/:token	MISLEADING	Same — 200, SPA shell, no review flow behind it
POST /api/reviews	NOT BUILT	Honest 404 — the /api guard answers correctly
GET /discovery/search	CUT	Geo discovery removed from scope
POST /consents, /passport/credentials	CUT	DigiLocker and consent artifacts cut

FIX NEEDED — unbuilt root routes return 200, unbuilt API routes return 404

/api/* paths that do not exist answer with an honest JSON 404. Root-level paths — /p/:handle and /r/:token — fall through to the single-page-app handler and return **200 with an HTML shell**. A judge scanning a QR would get a blank app, not an error. This asymmetry should be closed when those routes are built, and until then they should be treated as absent rather than pending.

5. Cross-cutting concerns

Concern	Status	Evidence
Authentication (§12.1)	PARTIAL	JWT spine complete: 15-min access, rotating refresh, reuse detection revoking the family, hashed OTP, attempt caps, throttling. Delivery stubbed; no login screen.
Authorisation (§12.1)	WORKS	Object-level, not role-only. Ownership is part of every query filter, so cross-user access returns 404 rather than leaking existence. Verified with two users.
Persistence (§7)	PARTIAL	5 of 15 collections in use: users, kaarigar_profiles, jobs, ledger_entries, otp_challenges, rate_bands. Cut collections are not stubs — they are out of scope.
Offline & sync (§9)	PARTIAL	Server half done: idempotent batch ingest, per-item reasons, retry classification. Client half absent: no outbox, no pending/synced/failed badges.
Validation (§6)	WORKS	Every write endpoint validates and names the offending field. Was zero at audit time.
Error handling (§9.2)	WORKS	The silent-failure defects are closed. Assistant responses carry a source discriminator; sync returns specific per-item reasons.
Rate limiting (§5)	PARTIAL	Assistant endpoint limited on uid <i>and</i> IP. In-process, so per-instance — must become shared before scaling.
Testing (§6)	PARTIAL	14 regression tests covering previously-fixed defects, plus a tree-completeness pre-commit hook. No unit tests, no CI.
Media / photos (§4A, §10)	CUT	No file input anywhere. Before/after photos removed from the demo loop.
Observability (§6)	NOT BUILT	Structured logging only. No Prometheus, Grafana or Sentry.

6. Things that must be fixed or decided

6.1 Rate-band figures are not sourced

The pricing engine is seeded with **32 placeholder bands that were invented, not taken from any published schedule**. Every band carries seededFrom = "PLACEHOLDER - not sourced, replace before demo", and the API returns that string as

the band's stated basis, so an unsourced band announces itself rather than passing as authoritative. **These must be replaced with real CPWD DSR / state PWD / minimum-wage figures before the demo.** §14.1 warns that claiming a source you do not have is the fastest way to lose a panel.

6.2 RateBand needs a unit field (frozen-contract change)

wageFloor is a daily wage, but tasks such as wall_paint_sqft are priced per square foot. The floor was raising Rs 8/sqft to Rs 350/sqft — roughly 25× market. An interim guard infers the unit from the task-code suffix and reports floorApplicable: false, but the correct fix is a unit field on RateBand. **types.ts is frozen; this needs all three tracks.**

6.3 The demo build ships a real credential

The assistant endpoint now requires a bearer token. Because no login screen exists, the demo build carries VITE_DEMO_TOKEN — a 12-hour token for the seeded demo worker, **confirmed present in dist/assets/*.js**. Anyone opening devtools on the demo build holds a working credential for that account. Acceptable against seed data; **must be deleted when login ships.**

6.4 Firebase phone OTP could not be made to deliver

Billing, provider, SMS region policy, authorized domains and API key were all confirmed correct. A valid 2,276-character reCAPTCHA token is produced and Google returns INVALID_APP_CREDENTIAL. Leading hypothesis, unconfirmed: backend provisioning lag on a new project. **The fallback OTP path is architecturally identical and only delivery is stubbed**, so switching back is a one-line change inside requireAuth. Retest is free and worth doing.

6.5 @types/react is not installed

Every .tsx file compiles with React types as implicit any. strictNullChecks is on and clean; full strict reports ~1,169 errors, almost all from this one omission rather than from real defects. Fixing it touches every component.

7. What stands between here and a presentable demo

#	Work	Owner	Why it matters
1	Public QR passport at /p/:handle, with a real scannable QR	Server-side	The audit's #1 gap and §1's stated differentiator vs Urban Company / JustDial. Currently an icon and a dead link.
2	Deploy (render.yaml is written and build-tested)	Track C	A QR encoding localhost is useless off the laptop. Also gates the Day 6 second-phone scan.
3	Login screen and AuthProvider	Track B	Removes the shipped demo credential and unblocks every authenticated screen.
4	Wire screens to the API behind USE_API	Track B	Seven endpoints are built and consumed by nothing. This is what makes data survive a device change.
5	Signed review link /r/:token and POST /reviews	Server-side	Completes the customer half of the demo loop.
6	Client outbox with pending/synced/failed badges	Track B	The offline beat §14.1 calls the one judges remember. Server side is ready.

The honest one-line summary

A judge looking at the running app today sees a polished multilingual interface with genuinely working AI voice extraction, backed by a substantial and well-tested API that the interface is not yet talking to. The two most quotable claims in the pitch — **the QR-verifiable passport** and **data that survives a device change** — are the two that cannot currently be demonstrated.

All statuses in this document were verified against a running server on commit 657049e. Endpoint results are live HTTP responses, not inferred from source.